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PF-Lab 2

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Class: BS-CS

Section: "E"

Assignment: Programming Fundamental

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Task #1

Type and save the following Programs in visual studio. Run these Programs & observe the output.

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a) Task #1

Input:-

```
#include <iostream>
Using namespace std;
int main ()
{
    char letter;
    letter = 'A';
    Cout << letter << '\n';
    letter = 'B';
    Cout << letter << '\n';
    Cin >> letter;
    return 0;
}
```

Outputs:-

```
A
B
```

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b)

Input:-

```
#include <iostream>
Using namespace std;
int main ()
{
    char letter;
    letter = 65;
    cout <<letter << '\n';
    letter = 66;
    cout << letter << '\n';
    cin >> letter;

    return 0;
}
```

output:-

65
66

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c)

Input:-

```
#include <iostream>
using namespace std;
int main()
{
  int letter;
  letter = 'A';
  cout << letter << '\n';
  letter = 'B';
  cout << letter << '\n';
  cin >> letter;
  return 0;
}
```

Output:-

A

B

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Task #2

Take two integers as input from the user apply arithmatic operations on them (+, -, *, /) as Print them on screen.

Input

```
1  #include <iostream>
2  using namespace std;
3  int main () {
4  int num1, num2;
5  cout<< "Enter first integer:";
6  cin >> num2;
7  cout << "Addition = "<< num1+ num2<< endl;
8  cout << substraction="<<num1-num2<<endl;
9  cout << Multiplication="<<num1*num2<<endl;
10 cout << Division = "num1/num2<< endl;
11 return 0;
12 }
```

Output

```
Enter first int: 20
Enter 2nd int : 73
Addition = 93
Substraction= -53
```

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Multipication= 1460

Divison= 0

[Program Finished]



Task#3

Write the output of the following code.

Input:-

```
#include<stdio.h>
#include<iostream>
Using namespace std;
int main(){
int val1;
int Val2;
int Val3;
Cout<<"Please enter your 3 number;;
Cin>> val1 >> val2>> Val3;
Cout<< val1 << "\n"<< val2<<"\n"Val3<<"\n";
cout<< val3<<"\n"<< Val2<<"\n"<< val1"\n";
return 0;
}
```

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Output:-

Please enter your 3 number: 10
40
50

40
50
10
40
50
50
40
10

[Program finished]

Task #4

Write a C++ Program to calculate the between two Point.

Note x1, y1, x2, y2 are all double value.

Formulae:-

$$d = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

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Inputs-

```
#include<iostream>
#include <cmath>
Using namespace std;

int main () {
 double x1,y1,x2,y2,d;
cout<< "Enter x1:";
 cin>> x1;
 cout<< "Enter y1:";
 cin>> y1;
cout<< "Enter x2:";
 cin >> x2
 cout<< "Enter y2:";
 cin>> y2;
 d= sqrt((x2-x1) * (x2-x1) +
          (y2-y1) * (y2-y1));
 cout<< "Distance=" << d << endl;
 return 0;
}
```

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outputs-

Enter x1: 2

Enter y1: 3

Enter x2: 5

Enter y2: 6

Distance = 4.24264

[Program finished]



Task #5

Calculate the area of a circle

(area = PI * r 2)

Calculate the area of a rectangle

(area = length * width)

Calculate the area of a triangle

(area = base * height * .5)

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Inputs-

```
#include <iostream>
using namespace std;
int main () {
    // 1. Area of Circle
    float r = 5;
    float circle = 3.14 * r * r;

    // 2. Area of Rectangle
    float length = 10, width = 5;
    float rectangle = length * width;

    // 3. Area of Triangle
    float base = 8, height = 6;
    float triangle = 0.5 * base * height;

    cout << " Area of circle = " << circle << endl;

    return 0;
}
```

outputs-

```
Area of Circle = 78.5
Area of Rectangle = 50
Area of Triangle = 24

[Program finished]
```

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Task #6

Total Purches A customer in a store is Purchasing five items. The Price of five items are:

Price of item 1 = $12.95

Price of item 2 = $24.95

Price of item 3 = $6.95

Price of item 4 = $14.95

Price of item 5 = $3.95

Write a Program that holds the Price of five items in five vairables. Display each items Price, the subtotal of sale, the amount of Sales tax, and the total. Assume the sales tax is 6%.

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Input:-

```
#include <iostream>
#include <iomanip>
using namespace std;
int main(){
    double item 1
    double item 2
    double item 3
    double item 4
    double item 5
  double subtotal;
double sales Tax;
double total;
Subtotal = item 1+ item 2 +item 3+ item 4 + item 5;
   Sales Tax= subtotal * 0.06;
    total= subtotal+Sales Tax;
   Cout << fixed << Setprecision (2);
   Cout << " Price of item 1 = $" << item 1 <<
   endl;
Cout<< "Price of item 2 = $" << item 2 << endl;
Cout<< "Price of item 3 = $" << item 3 << endl;
```

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```
cout << "Price of item 4 = $" << item4 << endl;
cout << "Price of item 5 = $" << item5 << endl;
cout << "subtotal = $" << subtotal << endl;
cout << " sales Tax = $" << salesTax << endl;
cout << "Total = $" << total << endl;
return 0;
}
```

outputs-

```
Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95
subtotal = $63.75
Sales Tax = $3.82
Total = $67.58
[Program finished]
```

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